PostgreSQL Setup Guide

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Retail Sales Analysis Project

A step-by-step tutorial for beginners

PostgreSQLpgAdmin 4SQLCSV Import

What you'll learn

- Install PostgreSQL and open pgAdmin.
- Create a database and a table.
- Import a CSV sales dataset.
- Execute SQL queries and inspect results.
- Troubleshoot common errors.

Chapter 1 — Download PostgreSQL



Download from the official website

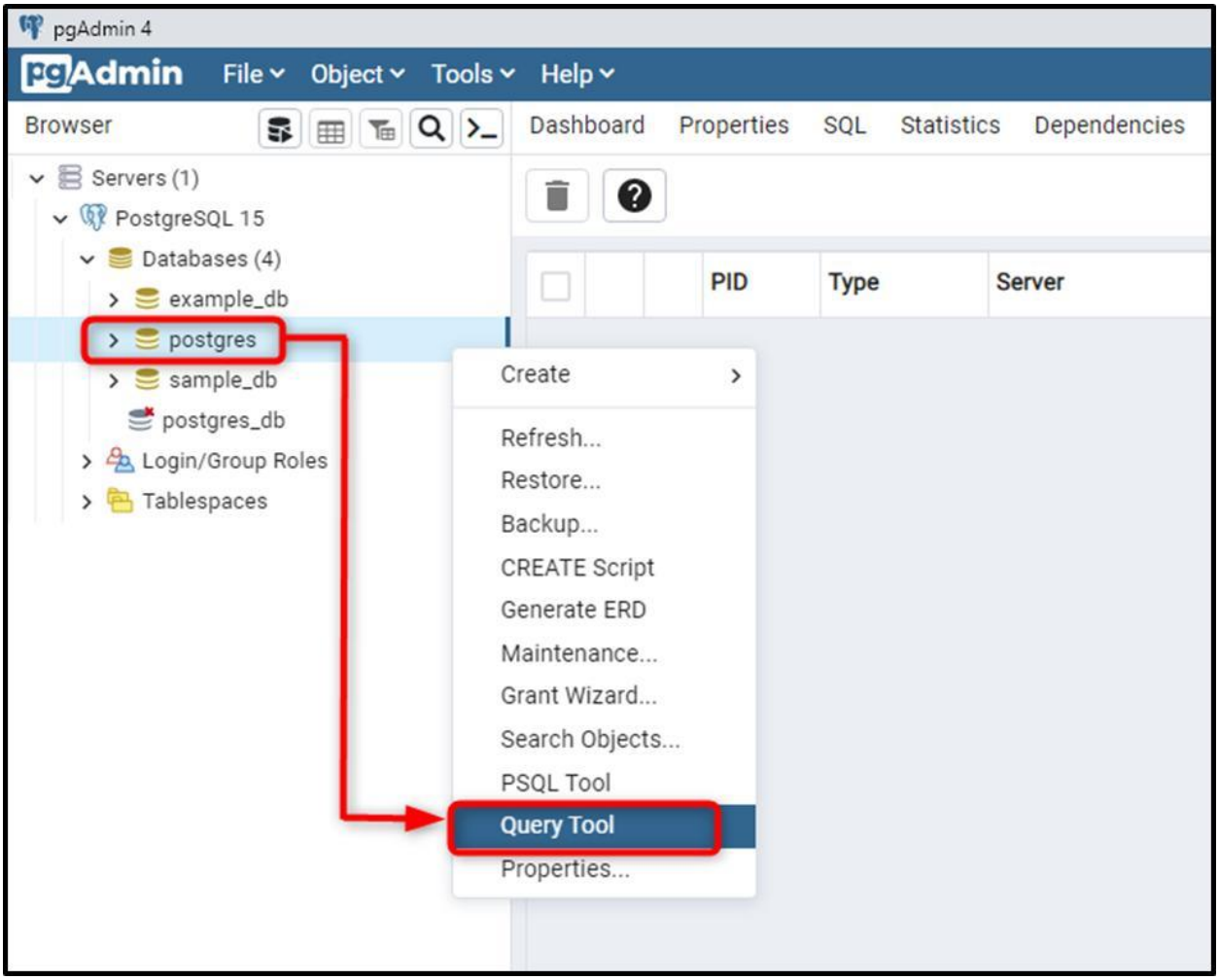
Visit the PostgreSQL Downloads page and select your operating system.

On Windows, use the official installer. Select PostgreSQL Server and pgAdmin 4 when available.

During installation:

1. Follow the installation wizard.
2. Set a password for the postgres administrative account.
3. Keep a note of the port number; the default is 5432.
4. Complete installation and launch pgAdmin 4.

Chapter 2 — Connect to PostgreSQL



1. Open pgAdmin 4.
2. Expand **Servers** in the Browser panel.
3. Select your PostgreSQL server.
4. Enter your password when prompted.

5. Confirm that the server connects successfully.

Typical local connection settings are:

Setting	**Default**
Host	localhost
Port	5432
Username	postgres

These values may differ on your computer.

Chapter 3 — Create the database

Open the Query Tool while connected to the maintenance database, such as postgres, and run:

sql

```
CREATE DATABASE retail_sales_db;
```

Execute this command separately. PostgreSQL does not permit CREATE DATABASE inside a transaction block.

Then connect to retail_sales_db in pgAdmin and open its Query Tool.

Chapter 4 — Create the sales table

Run the following SQL in the new database:

sql

```
CREATE TABLE retail_sales (
    transaction_id VARCHAR(50),
    transaction_date DATE,
    sales_channel VARCHAR(50),
    product_name VARCHAR(150),
```

```
  quantity INTEGER,
  unit_price NUMERIC(12, 2),
  discount_pct NUMERIC(5, 4)
);
```

This table is based on the assumed structure for your project.

Column	**Purpose**
transaction_id	Transaction or order identifier
transaction_date	Date of sale
sales_channel	Online or Physical Store
product_name	Product sold
quantity	Units sold
unit_price	Price before discount
discount_pct	Discount as a decimal

For example, 0.10 represents a 10% discount.

Chapter 5 — Prepare and import your CSV

Your source CSV should have compatible columns, such as:

csv

```
transaction_id,transaction_date,sales_channel,product_name,quantity,unit_price,discount_pct
ORD1001,2024-01-15,Online,Laptop,2,850.00,0.10
ORD1002,2024-01-16,Physical Store,Smartphone,3,600.00,0.05
```

ORD1003,2024-01-17,Online,Headphones,5,75.00,0.00

These rows are examples only, not your original dataset.

To import your file:

1. Expand retail_sales_db in pgAdmin.
2. Navigate to **Schemas → public → Tables**.
3. Right-click retail_sales.
4. Select **Import/Export Data**.
5. Choose **Import** and select your CSV.
6. Set the format to CSV and enable the header option if your first row contains column names.
7. Confirm the delimiter is a comma and check the column mapping.
8. Start the import and review the messages for errors.

Make sure you import into an empty table for your first attempt to avoid duplicate rows.

Chapter 6 — Verify your data

Check how many records were imported:

sql

```
SELECT COUNT(*) AS total_rows
FROM retail_sales;
```

Preview the data:

sql

```
SELECT *
FROM retail_sales
LIMIT 10;
```

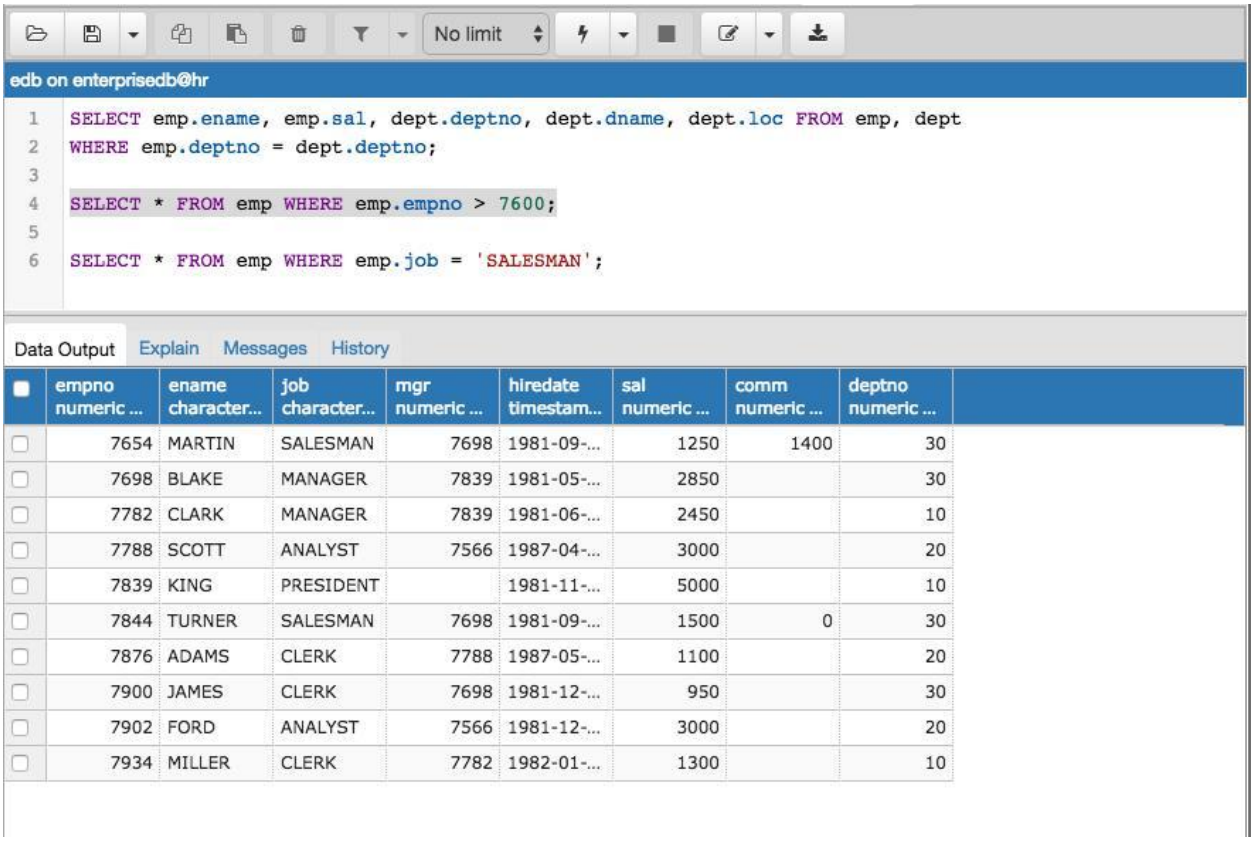
Check the date range:

sql

```
SELECT
  MIN(transaction_date) AS first_transaction,
  MAX(transaction_date) AS last_transaction
FROM retail_sales;
```

If these queries return sensible results, your data is ready for analysis.

Chapter 7 — Run your SQL analysis



1. Connect to retail_sales_db.
2. Open **Tools → Query Tool**.
3. Open your retail_sales_analysis.sql file.
4. Review the code and confirm that the table and column names match your dataset.
5. Execute the queries using the Query Tool's Execute control.

6. Review the **Data Output** and **Messages** panels.

Your project includes queries to analyze:

- Total revenue, order counts, and units sold.
- Revenue contribution by sales channel.
- Monthly revenue and month-over-month growth.
- Quarterly channel performance.
- Discount rates and revenue.
- Top-performing products.
- Product rankings and channel comparisons.

You can execute individual queries to review their results one at a time.

Chapter 8 — Understand the revenue calculation

The project assumes revenue is calculated as:

$$\text{Revenue} = \text{Quantity} \times \text{Unit Price} \times (1 - \text{Discount})$$

For two units priced at $100 each with a 10% discount:

$$2 \times 100 \times (1 - 0.10) = \$180$$

This formula assumes the discount is stored as a decimal and applies to the full unit price. If your dataset records discounts differently, adjust the formula before interpreting results.

Chapter 9 — Troubleshooting

Problem	**What to check**
Relation does not exist	Confirm the database, schema, and table name.
Invalid date syntax	Check the CSV date format.
Invalid numeric syntax	Remove currency symbols or percent signs from numeric fields.

Query returns no rows	Verify the table is populated and contains 2024 transactions.
Revenue differs from reported results	Check the dataset, discount convention, formula, and order definition.
CSV import fails	Review headers, column mapping, encoding, and data types.

Chapter 10 — Project completion checklist

Setup progress

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Install PostgreSQL and pgAdmin 4

Connect to the PostgreSQL server

Create retail_sales_db

Create the retail_sales table

Import the original CSV dataset

Verify row counts and data quality

Execute the SQL analysis queries

Validate the reported project findings

Upload the project files to GitHub

Final note

The original sales dataset has not been supplied, so the sample rows are illustrative. Verify the revenue contributions of 51.55% for Online and 48.45% for Physical Store, along with the 47.6% combined revenue contribution from Laptops and Smartphones, against your actual data before publishing your final results.